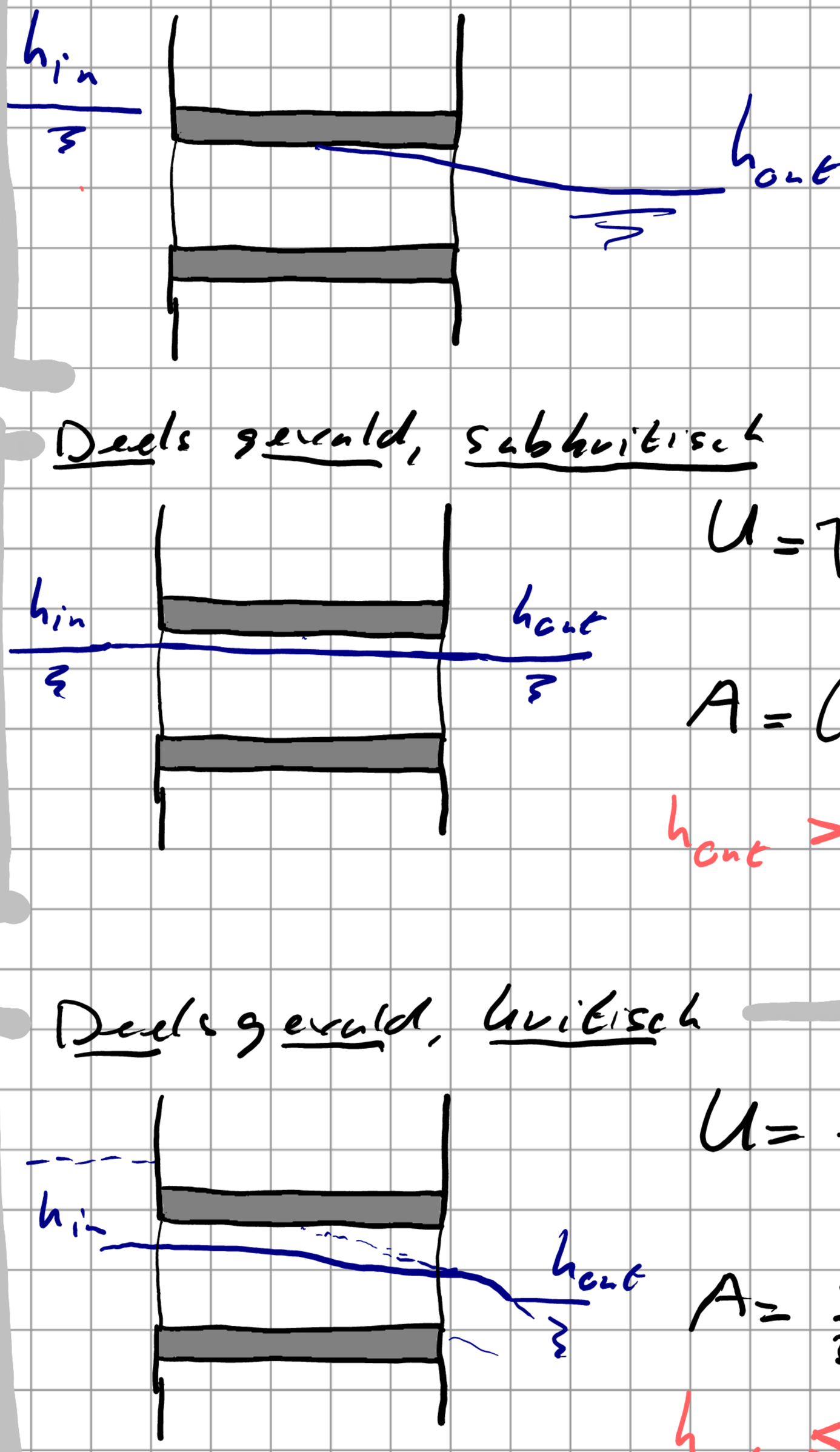
Afvoerverlatie Lochem

Definities



Deels gevuld, subkritisch

$$U = \sqrt{2g(h_{in} - h_{out})}$$

$$A = (h_{out} - h_{sill})\, B\, \mu$$

$$h_{out} > \frac{2}{3}(h_{in} - h_{sill}) \quad (1)$$

Deels gevuld, kritisch

$$U = \sqrt{2g\ \tfrac{1}{2}(h_{in} - h_{sill})}$$

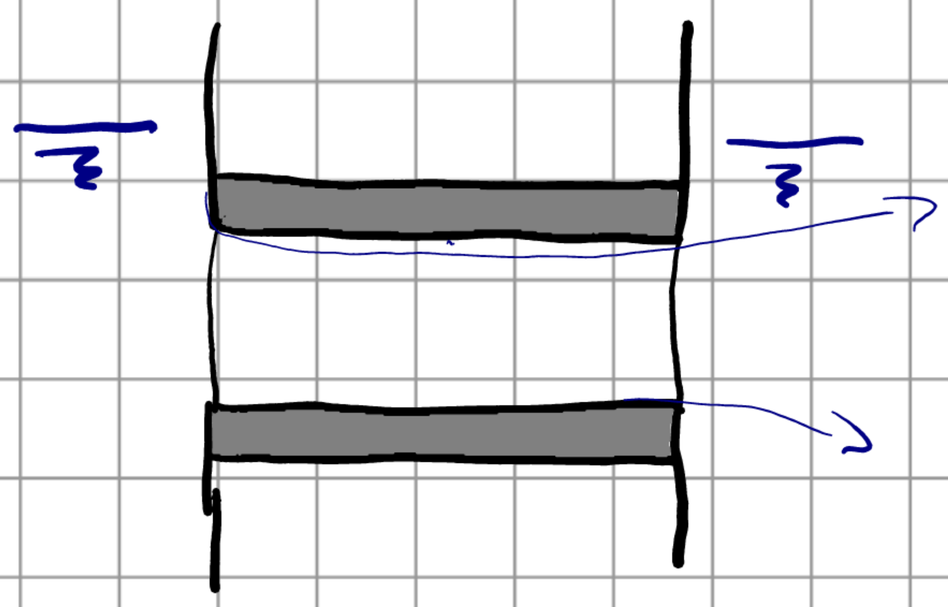
$$A = \frac{2}{3}(h_{in} - h_{sill})\, B\, \mu$$

$$h_{out} \leq \frac{2}{3}(h_{in} - h_{sill})$$

$$h_{cv} \leq h_{weir}$$

$$h_{cv} = \frac{2}{3}(h_{in} - h_{sill}) + h_{sill} = \frac{2h_{in} + h_{sill}}{3} \quad (2)$$

Gerald, subkritisch



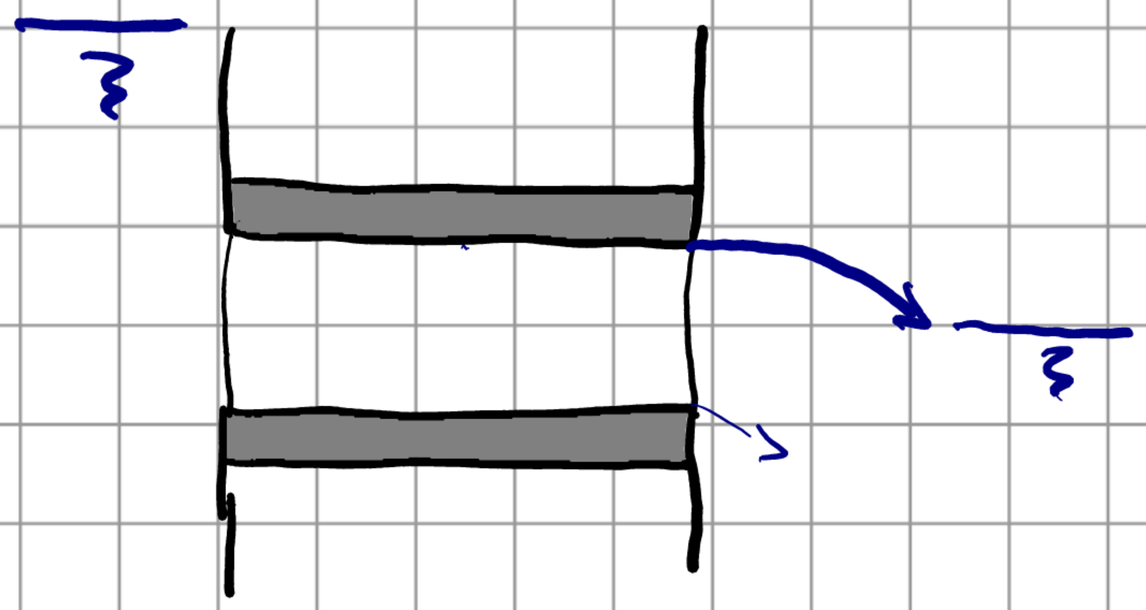
$$U = \sqrt{2g(h_{in} - h_{out})}$$

$$A = (h_{weir} - h_{sill})\, B/n$$

$$h_{out} > h_{weir}$$

(3)

Gerald, kritisch, gedrückte gestaut



$$U = \sqrt{2g(h_{in} - h_{weir})}$$

$$A = (h_{weir} - h_{sill})\, B/n$$

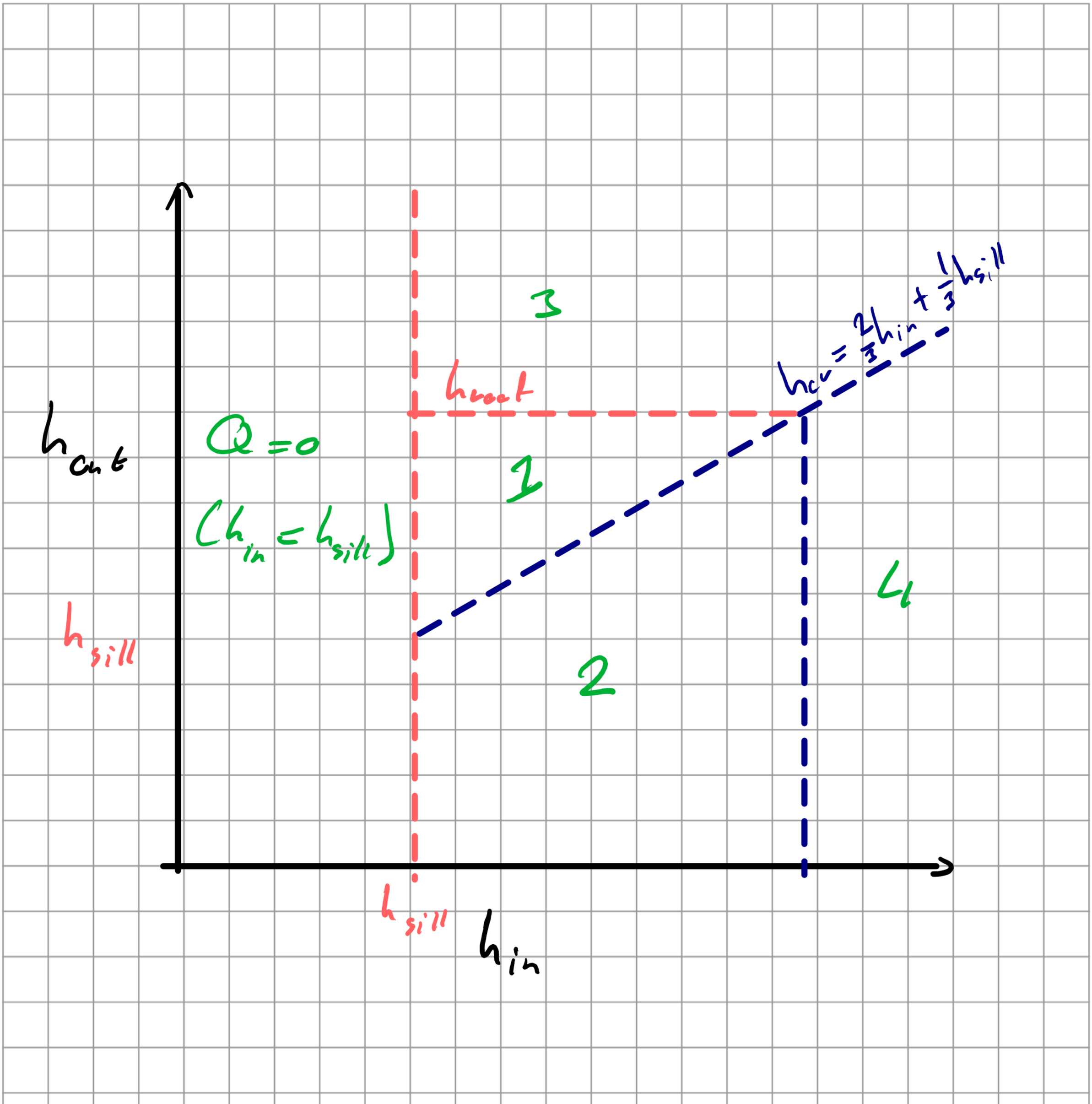
(4)

$$h_{out} \leq h_{cr}$$

$$h_{cr} \geq h_{weir}$$

Verzamelkabel cap. model

Cond.	Δh	d_{out}	
1	$h_{in} - h_{out}$	$h_{out} - h_{sill}$	$h_{out} > h_{cv}$ $h_{out} = h_{roof}$
2	$h_{in} - h_{cv}$	$h_{cv} - h_{sill}$	$h_{out} \equiv h_{cv}$ $h_{cv} \equiv h_{roof}$
3	$h_{in} - h_{out}$	$h_{roof} - h_{sill}$	$h_{out} \equiv h_{roof}$
L_1	$h_{in} - h_{roof}$	$h_{roof} - h_{sill}$	$h_{out} \equiv h_{cv}$ $h_{cv} \geq h_{roof}$


3
h_{crest}
$h_{cr} = \frac{2}{3} h_{in}$
$h_{cr} = \frac{1}{3} h_{sill}$
h_{out}
Q = 0
$(h_{in} = h_{sill})$
1
4
2
h_{sill}
h_{sill}
h_{in}

$h_{out} > h_{cv}$

$\Delta h = h_{in} - h_{out}$

$d = \min\begin{pmatrix} h_{out} \\ h_{roof} \end{pmatrix} - h_{sill}$

h_{out}

h_{sill}

$Q = 0$

$(h_{in} = h_{sill})$

$\Delta h = h_{in} - h_{out}$

$d = h_{roof} - h_{sill}$

h_{roof}

$h_{cv} = \frac{2}{3} h_{in} + \frac{1}{3} h_{sill}$

$\Delta h = h_{in} - h_{out}$

$d = h_{out} - h_{sill}$

$\Delta h = h_{in} - h_{roof}$

$d = h_{roof} - h_{sill}$

$\Delta h = h_{in} - h_{cv}$

$d = h_{cv} - h_{sill}$

h_{sill}

h_{in}

$h_{out} < h_{cv}$

$\Delta h = h_{in} - \min\begin{pmatrix} h_{cv} \\ h_{roof} \end{pmatrix}$

$\left(d = \min\begin{pmatrix} h_{cv} \\ h_{roof} \end{pmatrix} \right)$